SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_MN_Duluth.Intl.AP-Duluth.ANGB.727450_TMYx.epw
- Building Name: SmallOffice
- Building Location: Duluth, MN
- Total Floor Area: 511 m^2

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof	**Wall upgrade:** • Exterior wall insulation: Fiberglass Batts, target R-10.0 (skipped: target R-value already satisfied) **Roof upgrade:** • Roof insulation: Blown Fiberglass, target R-20.0

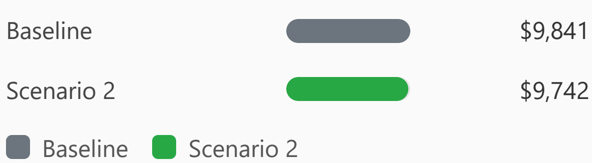
Scenario 2	wall + roof	**Wall upgrade:** • Exterior wall insulation: Fiberglass Batts, target R-13.0 (skipped: target R-value already satisfied) **Roof upgrade:** • Roof insulation: Blown Fiberglass, target R-24.4
Scenario 3	wall + roof	**Wall upgrade:** • Exterior wall insulation: Expanded Polystyrene (EPS) Foam Board, target R-13.0 (skipped: target R-value already satisfied) **Roof upgrade:** • Roof insulation: Extruded Polystyrene (XPS) Foam Board, target R-24.4

Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	236.54	234.60	**1.94**	**0.82%**
Scenario 2	Total Site Energy (GJ)	236.54	234.51	**2.03**	**0.86%**
Scenario 3	Total Site Energy (GJ)	236.54	234.51	**2.03**	**0.86%**

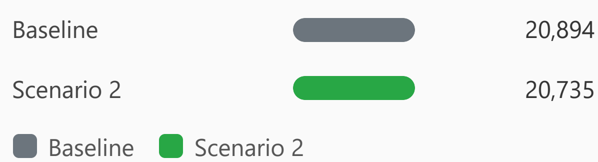
Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) ($/yr) Comparison



Max saving: **$-99 (-1.01%)**

Operational Carbon Saving (Scenario - Baseline) (kg CO_2e/yr) Comparison



Max saving: **-159 kg CO_2e (-0.76%)**

Min Retrofit Construction Cost

$5,241

Scenario 1

Min Retrofit Embodied Carbon

5,551 kg CO_2e

Embodied carbon intensity (ECI): 11 $kgCO_2e/m^2$

Scenario 1

Lowest Retrofit Cost Payback Period

58 years

Scenario 1

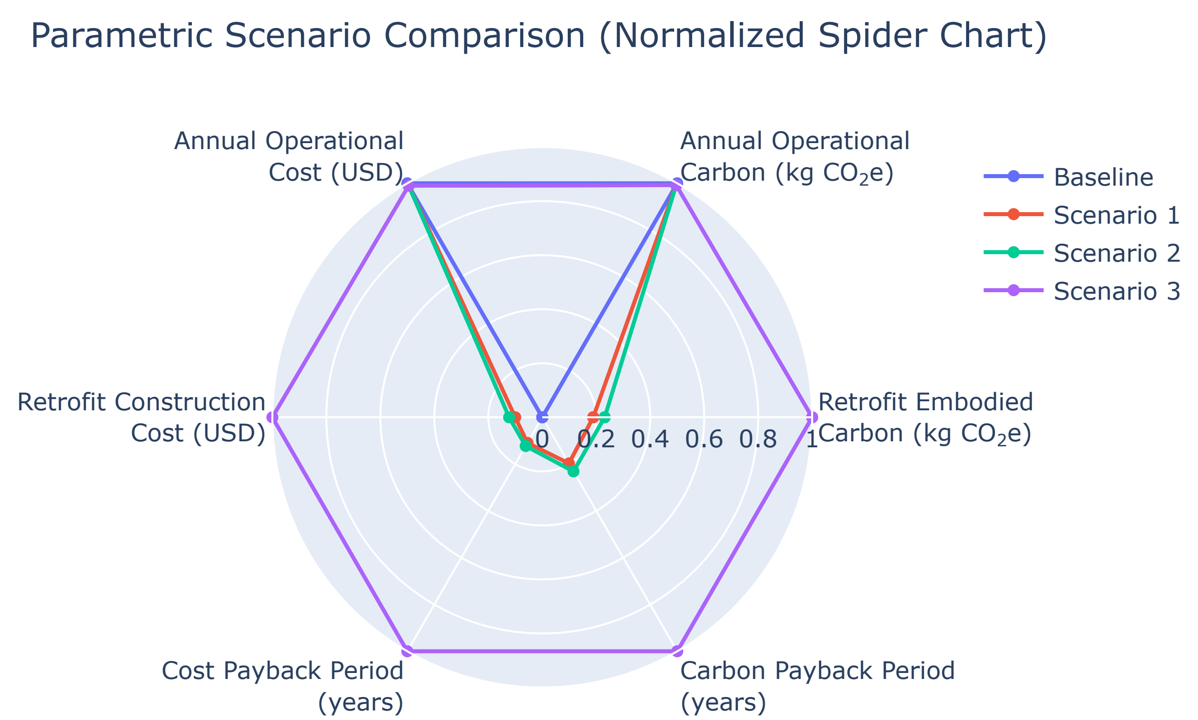
Lowest Retrofit Carbon Payback Period

36 years

Scenario 1

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) ($/yr)).

Scenario 1	58 yrs
Scenario 2	65 yrs
Scenario 3	526 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO_2e/yr)).

Scenario 1	36 yrs
Scenario 2	43 yrs
Scenario 3	185 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

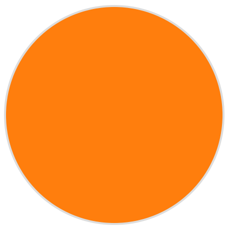
Scenario	Component	Material name	Volume (m^3)	Area (m^2)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m^3)	Product Lifetime (years)
Scenario 1	Wall insulation	Fiberglass Batts	N/A	N/A	N/A	N/A	0.03	30	60
Scenario 1	Roof insulation	Blown Fiberglass	82	599	0.14	N/A	0.03	30	68
Scenario 2	Wall insulation	Fiberglass Batts	N/A	N/A	N/A	N/A	0.03	30	60
Scenario 2	Roof insulation	Blown Fiberglass	101	599	0.17	N/A	0.03	30	68
Scenario 3	Wall insulation	Expanded Polystyrene (EPS) Foam Board	N/A	N/A	N/A	N/A	0.03	20	60
Scenario 3	Roof insulation	Extruded Polystyrene (XPS) Foam Board	101	599	0.17	N/A	0.03	26	75

Result Summary

> Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

Scenario 1

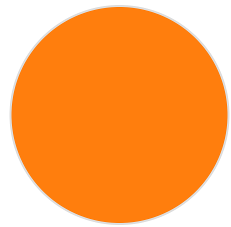
Cost breakdown by retrofit measure



Roof: $5,241 (100.0%)

Total: $5,241

Carbon breakdown by retrofit measure

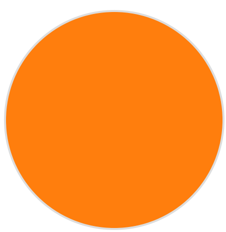


Roof: 5,551 (100.0%)

Total: 5,551 kg CO_2e

Scenario 2

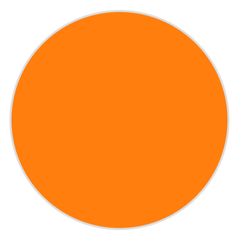
Cost breakdown by retrofit measure



Roof: $6,420 (100.0%)

Total: $6,420

Carbon breakdown by retrofit measure

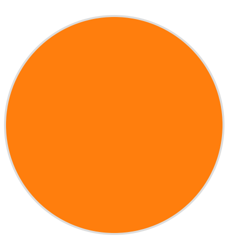


Roof: 6,801 (100.0%)

Total: 6,801 kg CO_2e

Scenario 3

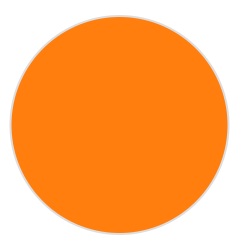
Cost breakdown by retrofit measure



Roof: $52,261 (100.0%)

Total: $52,261

Carbon breakdown by retrofit measure



Roof: 29,412 (100.0%)

Total: 29,412 kg CO_2e

Scenario	Retrofit Embodied Carbon (kg CO_2e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO_2e)	Annual Operational Cost (USD)
Baseline	0.00	$0.00	20,894	$9,841
Scenario 1	5,551	$5,241	20,742	$9,750
Scenario 2	6,801	$6,420	20,735	$9,742
Scenario 3	29,412	$52,261	20,735	$9,742

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